

Surface Pond Environment Model

LIFEORIG working model specification

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1 Purpose

The surface pond environment is a shallow liquid reactor exposed to atmosphere, radiation, sediment or mineral surfaces, and changing liquid level. It is different from a hydrothermal vent and different from volcanic rock pores. The liquid does not have to be water: the same class should represent an aqueous pond, a brine, a hydrocarbon pond, or a custom exoplanetary liquid mixture.

The key physical idea is:

shallow pond $\rightarrow$ liquid-level cycles $\rightarrow$ concentration/dilution $\rightarrow$ UV and sediment chemistry

(1)

The first implementation should create the physical environment object. It may initially use a single well-mixed pond approximation, but the object must be organized so that hydrodynamic, radiative, and transport structure can be added without changing the input parser.

2 Geometry

The first surface pond model is zero-dimensional: one well-mixed shallow liquid body above a sediment or mineral surface. A later model should allow a depth-averaged horizontal velocity field and vertical profiles of radiation, temperature, and chemistry.

```
atmosphere / radiation input
|
v
+-----+
| shallow liquid          |
| dissolved molecules     |
+-----+
| sediment / mineral floor |
+-----+
```

The geometry is defined by:

$$V(t) = A_{\text{pond}} h(t), \quad (2)$$

where $V(t)$ is liquid volume, A_{pond} is pond area, and $h(t)$ is liquid level.

3 State variables

Symbol	Code name	Meaning
t	<code>time</code>	simulation time
A_{pond}	<code>area</code>	horizontal pond area
$h(t)$	<code>liquid_level</code>	instantaneous liquid level
$V(t)$	<code>volume</code>	liquid volume, $A_{\text{pond}}h(t)$
T	<code>temperature</code>	pond water temperature
P	<code>pressure</code>	local pressure
pH	<code>pH</code>	optional aqueous acidity/basicity; not defined for all liquids
I	<code>ionic_strength</code>	optional aqueous/electrolyte salt strength
F_{λ}	<code>radiation_field</code>	wavelength-dependent radiation forcing
$\mathbf{u}$	<code>velocity_field</code>	fluid velocity, initially optional
g	<code>gravity</code>	local gravitational acceleration
A_{sed}	<code>sediment_surface_area</code>	reactive sediment/mineral area
C_i	<code>concentrations[species]</code>	concentration of species i

4 Liquid-level cycle

The simplest liquid-level cycle is periodic:

$$h(t) = h_0 + \Delta h \cos\left(\frac{2\pi t}{\tau_{\text{cyc}}}\right), \quad (3)$$

with clipping:

$$h(t) \geq h_{\min}. \quad (4)$$

The variables mean:

- h_0 is the mean liquid level.
- Δh is the amplitude of drying/refilling.
- τ_{cyc} is the cycle period.
- $h_{\min}$ prevents the pond from becoming mathematically negative.

5 Concentration and dilution

When the pond dries, the same number of molecules occupies less liquid. The concentration rises. When the pond refills, concentration falls.

If h_{old} is the previous depth and h_{new} is the new depth:

$$C_i^{\text{new}} = C_i^{\text{old}} \frac{h_{\text{old}}}{h_{\text{new}}}. \quad (5)$$

This is the central surface-pond mechanism. It is why shallow ponds can drive prebiotic chemistry: second-order and higher-order reactions speed up strongly when concentrations increase during drying.

6 Hydrodynamic Layer

The final surface pond model should not hard-code a single chemical equation at the start. It should first define the physical transport layer that later chemical, coacervate, and protocell equations will couple to. The continuum variables are:

$$\rho, \quad \eta, \quad \nu = \eta/\rho, \quad \kappa_T, \quad D_i, \quad \mathbf{u}(\mathbf{x}, t), \quad p(\mathbf{x}, t), \quad T(\mathbf{x}, t), \quad (6)$$

where ρ is liquid density, η dynamic viscosity, ν kinematic viscosity, κ_T thermal diffusivity, D_i molecular diffusivity, and $\mathbf{u}$ the liquid velocity.

For a Newtonian liquid, the reference hydrodynamic model is incompressible Navier–Stokes with buoyancy and optional interfacial forcing:

$$\nabla \cdot \mathbf{u} = 0, \quad (7)$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \eta \nabla^2 \mathbf{u} + \rho \mathbf{g} + \mathbf{f}_{\text{surf}} + \mathbf{f}_{\text{ext}}. \quad (8)$$

Here $\mathbf{f}_{\text{surf}}$ represents surface-tension-gradient forcing (Marangoni stresses), and $\mathbf{f}_{\text{ext}}$ represents later model terms such as wind stress, rainfall impact, or externally imposed shaking.

For shallow ponds, the first useful reduction is depth-averaged transport:

$$\frac{\partial h}{\partial t} + \nabla_{\parallel} \cdot (h \mathbf{U}) = R_{\text{rain}}(t) - E_{\text{evap}}(t) - Q_{\text{runoff}}, \quad (9)$$

where $\mathbf{U}(x, y, t)$ is the depth-averaged horizontal velocity. The zero-dimensional implementation is the special case $\mathbf{U} = 0$ and $h = h(t)$.

6.1 Convection and internal heat

If the planet supplies an internal heat flux, or if the surface is heated or cooled by radiation, the pond may develop buoyancy-driven convection. A first diagnostic is the Rayleigh number:

$$\text{Ra} = \frac{g \beta \Delta T h^3}{\nu \kappa_T}, \quad (10)$$

where β is the thermal expansion coefficient and ΔT the vertical temperature contrast. For a simple layer, convection begins when Ra exceeds a critical value of order 10^3 , with the classical rigid/free-boundary number depending on boundary conditions. This should be used as a switch for whether the pond is treated as purely diffusive, well-mixed, or explicitly convective.

Thermal forcing should therefore be represented by a model object:

$$\mathcal{T} = \mathcal{T}(F_{\lambda}^*, \tau_{\text{atm}}, q_{\text{int}}, h, \rho, c_p, \kappa_T), \quad (11)$$

where F_{λ}^* is the stellar spectrum at the planet, τ_{atm} atmospheric optical depth, and q_{int} internal heat flux from the planetary body.

6.2 Surface tension and evaporation-driven flow

Surface ponds and droplets can also be moved by gradients in surface tension:

$$\boldsymbol{\tau}_{\text{interface}} = \nabla_s \gamma(T, \mathbf{C}), \quad (12)$$

where γ is surface tension and ∇_s is the surface gradient. This is the Marangoni mechanism. It matters for evaporating aqueous droplets, hydrocarbon liquids, and chemically active protocell-like droplets. Evaporation also creates capillary replenishment flow toward pinned contact lines:

$$\mathbf{U}_{\text{evap}} \sim \frac{J_{\text{evap}} L}{\rho h}, \quad (13)$$

where J_{evap} is evaporative mass flux and L a horizontal length scale. This transport can concentrate or segregate suspended particles, polymers, coacervates, and protocell candidates.

6.3 Generic chemical transport equation

The chemical equations should be finalized only after the hydrodynamic and compartment stage is chosen. The general environmental transport form is:

$$\frac{\partial C_i}{\partial t} + \nabla \cdot (C_i \mathbf{u}) = \nabla \cdot (D_i \nabla C_i) + R_i^{\text{env}}(\mathbf{C}, T, \text{solvent}, F_\lambda) + S_i - L_i. \quad (14)$$

The current well-mixed equation is the volume-averaged limit of this expression.

7 Compartment Stage Coupling

Different protocell-formation stages should not all obey the same equation of motion. The surface pond environment supplies fields

$$\mathbf{u}, \quad T, \quad F_\lambda, \quad h, \quad \rho, \quad \eta, \quad D_i, \quad A_{\text{reactive}}, \quad (15)$$

and each compartment class uses those fields differently.

7.1 Dissolved chemicals

Dissolved species follow advection–diffusion–reaction:

$$\partial_t C_i + \nabla \cdot (C_i \mathbf{u}) = \nabla \cdot (D_i \nabla C_i) + R_i. \quad (16)$$

7.2 Coacervate or dense liquid droplets

Coacervates should be treated as deformable dense droplets or particles with growth, fusion, breakup, and hydrodynamic transport:

$$\frac{d\mathbf{x}_k}{dt} = \mathbf{u}(\mathbf{x}_k, t) + \mathbf{v}_{\text{drift}}^{(k)} + \sqrt{2D_k} \boldsymbol{\xi}_k(t), \quad (17)$$

$$\frac{dV_k}{dt} = J_{\text{in}}^{(k)} - J_{\text{out}}^{(k)} + R_{\text{phase}}^{(k)}. \quad (18)$$

The drift term may include buoyancy, sedimentation, thermophoresis, Marangoni motion, or chemical taxis.

7.3 Vesicles and protocells

Vesicles and protocells add a boundary state. Their external motion can still be advective/diffusive, but their internal chemistry obeys exchange through a structured interface:

$$\frac{dC_{i,k}^{\text{in}}}{dt} = R_{i,k}^{\text{in}} + \frac{A_k}{V_k} P_{i,k} (C_i^{\text{ext}}(\mathbf{x}_k, t) - C_{i,k}^{\text{in}}) - D_{i,k}^{\text{damage}}. \quad (19)$$

Thus coacervates, vesicles, and protocells all live inside the same pond environment, but they do not share the same state equations.

8 Chemical update

The minimal concentration equation is:

$$\frac{dC_i}{dt} = R_i(\mathbf{C}, T, \text{solvent}, F_\lambda, A_{\text{sed}}) + S_i^{\text{atm}} - k_i^{\text{photo}} F_\lambda C_i - k_{\text{wash}} C_i - \frac{C_i}{h} \frac{dh}{dt}. \quad (20)$$

The terms mean:

- R_i is production or loss from the chemical reaction network.
- S_i^{atm} is atmospheric input or deposition of species i .
- $k_i^{\text{photo}} F_\lambda C_i$ is photolysis loss.
- $k_{\text{wash}} C_i$ is washout or runoff loss.
- $-(C_i/h)(dh/dt)$ is concentration during drying or dilution during refilling.

9 Code structure

The first implementation is:

SurfacePond

```
state: PondState
concentrations
atmospheric_input
photolysis_rates
liquid_level(t)
```

PondState

```
time
liquid_level
temperature
pressure
pH optional
ionic_strength
radiation_field
gravity
sediment_surface_area
volume
concentrations
```

10 Comparison with other environments

Environment	Main driver	Main geometry
Surface pond	radiation plus liquid-level cycles, flow, and concentration	shallow liquid body
Volcanic rock	wet/dry films and rock pores	porous surface rock/interstices
Hydrothermal vent	temperature, pH, redox, and flow gradients	porous chimney wall

11 First implementation target

The first target is not a full ecological model. It is:

1. hold a shallow pond state;
2. update liquid level in time;
3. concentrate or dilute chemical species as liquid level changes;
4. add atmospheric source terms;
5. apply radiation-driven photolysis and washout losses;
6. expose the state to the chemical reaction network.

12 Hydrodynamics References

The hydrodynamic layer should be based on established continuum fluid mechanics and then reduced to the simplest model needed for each environment. Core references are:

- Batchelor’s *An Introduction to Fluid Dynamics* for incompressible flow and transport.
- Landau and Lifshitz, *Fluid Mechanics*, for continuum equations, interfacial stresses, and transport.
- Chandrasekhar’s *Hydrodynamic and Hydromagnetic Stability* for Rayleigh–Benard convection and instability thresholds.
- Pearson’s analysis of surface-tension-driven convection for Marangoni cells.
- Scriven and Sternling for Marangoni effects at fluid interfaces.
- Deegan et al. and Hu–Larson for evaporation-driven capillary and Marangoni flows in drying droplets.
- Zwicker et al. and Hanczyc et al. for chemically active droplets and protocell-like self-propelled liquid systems.

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